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Project on -
Need for a High-Performance Game Analytics Platform

Briefing

The client operates a CLOUD GAMING SERVICE , delivering high-performance gaming experiences through streaming. They manage vast amounts of data generated by users, including gameplay interactions and system metrics, and require efficient data processing, real-time analysis, and robust security to ensure smooth gameplay and optimal user engagement.

Challenges

- Difficulty querying and analyzing massive gaming dataset.
- Trouble in quickly processing big datasets for real-time decisions.
- Worries about managing data security and who can access it.
- Challenges in coordinating different systems across multiple servers.
- Problems with automating and scheduling important data tasks.
- Issues with storing and handling large amounts of data efficiently.

Consequence

- **Performance Issues**: As the number of players grew, the system encountered significant performance bottlenecks, leading to slower game performance and reduced player satisfaction.

- **Revenue Impact:** The lack of real-time insights into player activity affected the ability to monetize effectively, leading to a negative impact on **revenue**.

- **Understanding Player Preferences:** Analyzing player behavior and preferences was challenging, resulting in missed opportunities to enhance in-game experiences.

Solution

To address the numerous challenges faced by our client, we delivered a range of tailored services as comprehensive solutions.

- AWS S3 for Data storage.
- AWS Athena for Data Querying.
- AWS Kinesis for real-time ingestion of Logs and events.
- AWS EMR for large-scale batch and streaming processing with spark on EMR.
- AWS IAM for Data Security and Access Control.
- AWS Glue for automated ETL pipelines and maintaining a centralized data catalog.
- AWS CloudWatch for monitoring performance and setting up real-time alerts.

Impact

- **Subscription Upgrades:** A 19% increase in players opting for premium subscriptions.
- **Campaign Scalability:** Ability to run 7 times more marketing campaigns simultaneously.
- **Personalized Gaming:** Refined targeting algorithms elevated player engagement and satisfaction.
- **Predictive Analytics:** Enhanced algorithms identify player goals, increasing in-game transactions.
- **User Research:** Deeper insights into player behavior, improving game development.

Cluster & Capacity Planning

- **Daily Ingestion Volume: 40 GB/day**
- **Average Throughput: ~0.5 MB/sec** (40 GB ÷ 86,400 sec).
- **Peak Throughput: ~2.5 MB/sec** (Assuming 5x spike during game launches).
- **Event Rate: ~500 - 2,500 events/sec** (Assuming avg log size of 1KB).
- **Total Data Stored: ~35 TB** (20 TB Historical + 15 TB Yearly Growth).
- **Storage Strategy:**

1. **S3 Raw Zone:**
   Volume: 40 GB/day.
   Cost Efficiency: Since 40GB is small, we use **S3 Standard** for the first 30 days, then **Intelligent-Tiering** moves it to Infrequent Access (IA) automatically.

2. **S3 Processed Zone:**
   Volume: ~5 GB/day (Aggregated/Cleaned Parquet is usually 10-15% of raw JSON size).

Cluster & Capacity Planning

KINESIS CAPACITY:

- 1 Shard = 1 MB/sec write capacity.
- Throughput: ~0.5 MB/sec (Automatically handled by On-Demand Mode.
- Retention: 7 days (not default 24 hrs) for replay capability

S3 STORAGE:

- **Raw Zone: 40 GB/day → 14 TB/year**
- **Processed Zone: 5 GB/day (after aggregation)**
- **S3 Intelligent-Tiering moves data to IA after 30 days.**
- **Est. Storage Cost: ~$100 - $150 / month (Very affordable).**

EMR CLUSTER:

- **Master: 1x m5.xlarge (On-Demand)**
- **Core: 2x m5.xlarge (On-Demand)**
- **Task:** 2x **m5.xlarge** (Spot Instances)

ATHENA:

- **Estimated scans: 1-2 TB/week → ~$5-10/week in query costs.**
- **Partition Projection reduces query planning latency by 90%**

AWS SERVICES USED

- **Ingestion Layer :** Kinesis Data Streams (On-Demand), Kinesis Data Firehose
- **Storage Layer:** S3 (Standard → Intelligent-Tiering), Glue Data Catalog
- **Processing Layer:** EMR (Transient Cluster + Spark), Spot Instances (Cost Optimization)
- **Analytics Layer:** Athena (Partition Projection for speed), Glue Crawler (Schema Discovery)
- **Visualization:** QuickSight (SPICE for cost, Direct Query as fallback)
- **Security & Ops:** IAM (Role-based), VPC Endpoints (S3 cost savings), CloudWatch (Monitoring).

FLOW DIAGRAM

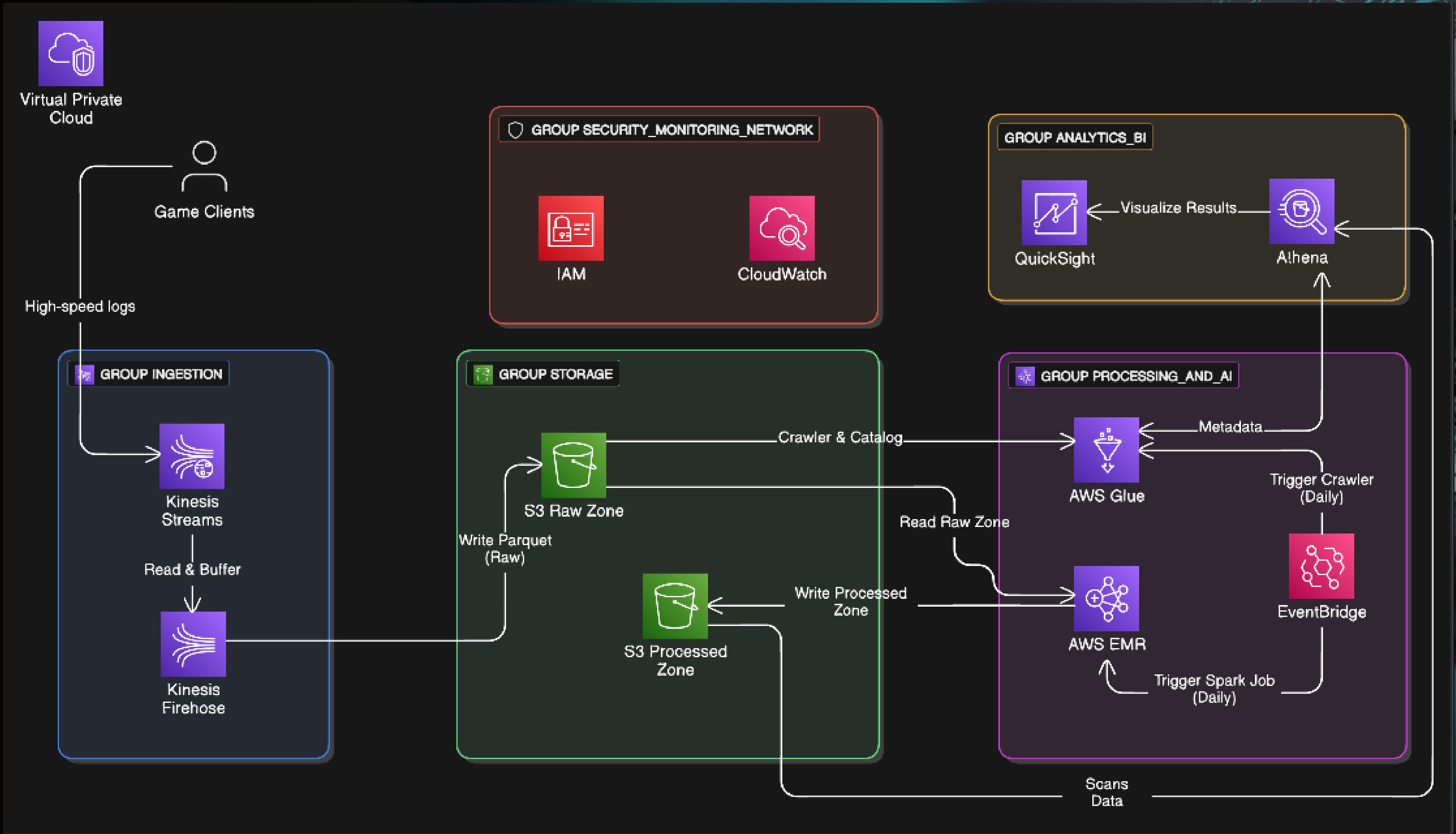
It shows the end-to-end data pipeline:

- Game Clients → Kinesis Data Streams → Kinesis Firehose → S3 Raw Zone (Parquet) → Glue Crawler → Glue Data Catalog → [Split into TWO paths]

  1. Real-Time Path (Future): → Lambda (Fraud Detection)
  2. Batch Path: → EMR (Spark - Deduplication + Aggregation) → S3 Processed Zone → Athena → QuickSight“

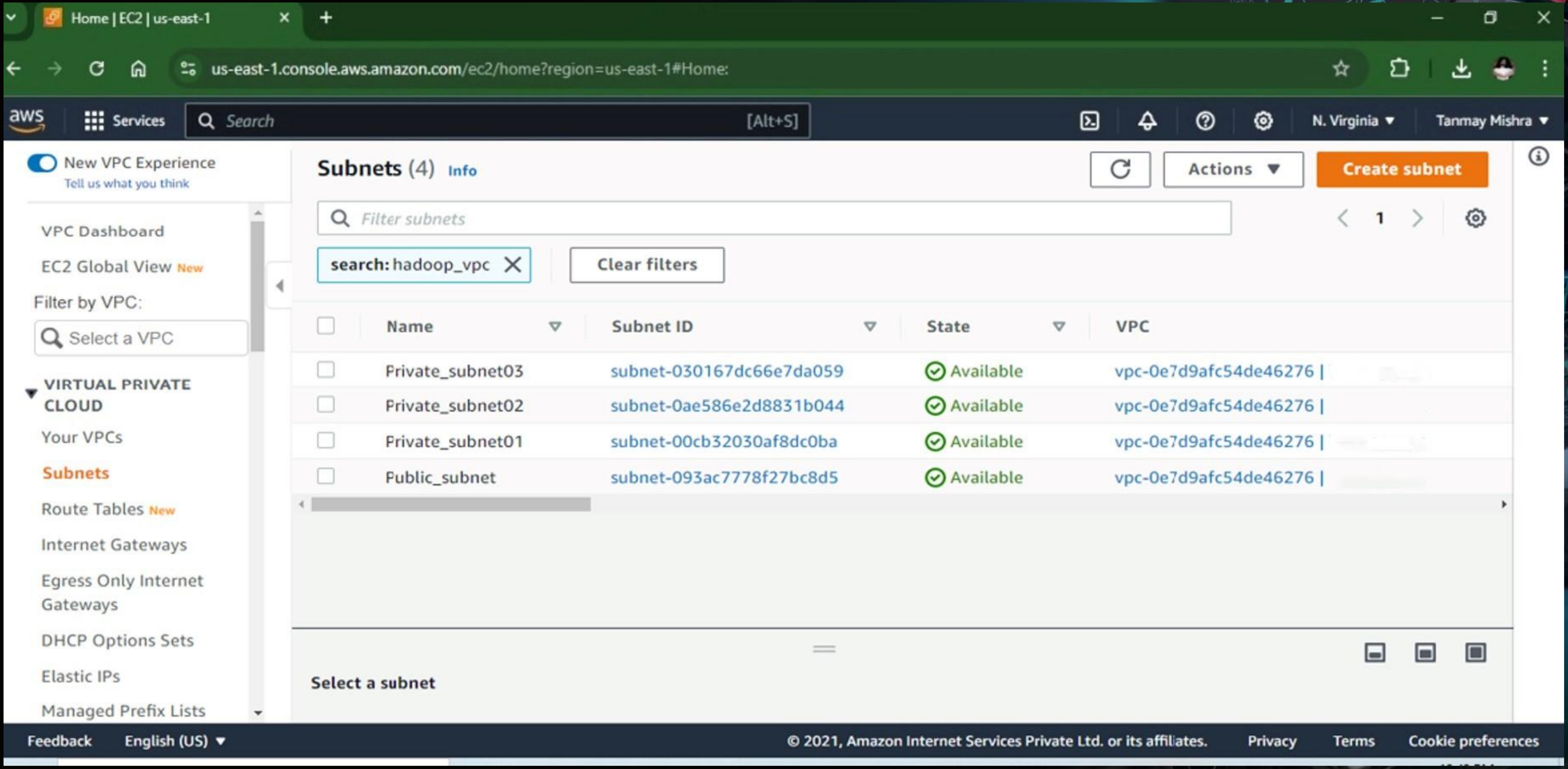
- With supporting services: IAM (Roles for each service), CloudWatch (IteratorAge, DPU metrics), VPC Endpoints (S3 access from private subnets)

ARCHITECTURE



VPC

- Deployed workloads inside secure VPC with private/public subnets.
- NAT Gateway enabled controlled internet access for private subnets.



IAM Roles

1. **Kinesis Producer Role (for game clients)**
   - Permissions: kinesis:PutRecords
   - Assumed by Game Server/Client via Cognito or Web Identity Federation (STS)
2. **Firehose Role**
   - Permissions: s3:PutObject (for Raw Zone)
   - Permissions: glue:GetTableVersion (for Parquet schema)
3. **EMR Cluster Role**
   - Permissions: s3:GetObject (Raw Zone), s3:PutObject (Processed Zone)
   - Permissions: glue:GetDatabase, glue:GetTable (for Glue Catalog) - Permissions: cloudwatch:PutMetricData (emit custom metrics)
4. **Athena Role**
   - Permissions: s3:GetObject (Processed Zone)
   - Permissions: glue:GetTable, glue:GetPartitions (read catalog)
5. **QuickSight Role**
   - Permissions: athena:StartQueryExecution (if Live)
   - OR: s3:GetObject (SPICE mode, one-time import)

Overview

- Using AWS for cloud-based high-graphics gaming, the company enhanced operational efficiency and user experience.
- AWS's real-time analytics identified performance bottlenecks, allowing quick adjustments that improved gaming performance.
- Additionally, AWS enabled precise monitoring of gaming transactions, quickly spotting unusual data usage and fraud, thus reducing fraudulent activities that previously went undetected.

END